

Ph.D. Exam: Numerical Analysis, May, 2025  
 Do 4 (four) of the first 5 (1-5) and 4 (four) of the  
 last 5 problems (6-10).

1. Assume  $A \in C^{m \times m}$ .

- (a) Show that  $A$  has a Schur decomposition.
- (b) If  $A$  is normal, show that  $A$  is diagonalizable.

2. Suppose  $A$  is Hermitian positive definite.

- (a) Prove that each principal submatrix of  $A$  is Hermitian positive definite.
- (b) Prove that an element of  $A$  with largest magnitude lies on the diagonal.
- (c) Prove that  $A$  has a Cholesky decomposition.

3. (a) Show that  $\|x\|_\infty$  is equivalent to  $\|x\|_2$  for all  $x \in R^n$ . That is to find  $C$  and  $c$  such that  $c\|x\|_\infty \leq \|x\|_2 \leq C\|x\|_\infty$ , for all  $x \in R^n$ . Note that the constants should be determined so that the equalities hold for some nonzero  $x \in R^n$ .

(b) Show that  $\|QA\|_2 = \|A\|_2$  if  $Q$  is a unitary matrix.

4. Assume that  $A \in C^{n \times n}$  and there exists  $p \geq 1$  such that  $\|A\|_p < 1$ , where  $\|\cdot\|_p$  is a vector-induced matrix norm.

- (a) Prove that  $I - A$  is invertible.
- (b) Prove that

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k.$$

(c) Prove that  $\|A\|_q \|A^{-1}\|_q \geq 1$ ,  $\forall 1 \leq q \leq \infty$ .

(d) Prove that

$$\frac{1}{1 + \|A\|_p} \leq \|(I - A)^{-1}\|_p \leq \frac{1}{1 - \|A\|_p}.$$

5. Let  $A = U\Sigma V^*$  be the singular value decomposition of  $A \in C^{m \times n}$ . Let  $u_j$  denote column  $j$  of  $U$ .

(a) Suppose  $\text{rank}(A)=p < n < m$ . Show  $\{u_1, u_2, \dots, u_p\}$  is a basis for  $\text{Col}(A)$  and  $\{u_{p+1}, u_{p+2}, \dots, u_m\}$  is a basis for  $\text{Null}(A^*)$ .

(b) Suppose  $A$  is full rank and  $x \neq 0$ . Let  $\sigma_i, i = 1, \dots, n$  be the singular values of  $A$ . Show

$$\sigma_1 \geq \frac{\|Ax\|_2}{\|x\|_2} \geq \sigma_n > 0.$$

If you want to use the property that  $\|A\|_2 = \sigma_1$ , then you must prove that it holds.

6. Consider the function  $g(x) = e^{-x}$ .

(a) Prove that  $g$  is a contraction on  $G = [\ln 1.1, \ln 3]$ .

(b) Prove that  $g$  maps  $G = [\ln 1.1, \ln 3]$  into  $G = [\ln 1.1, \ln 3]$ .

(c) Prove that  $x_{k+1} = g(x_k)$  converges to a unique fixed point  $z \in G = [\ln 1.1, \ln 3]$  for any initial value  $x_0 \in G = [\ln 1.1, \ln 3]$ .

7. Based on  $u_1(x) = 1, u_2(x) = x, u_3(x) = x^2$ , use Gram-Schmidt orthogonalization process to compute the three polynomials  $w_1(x), w_2(x), w_3(x)$  which are orthonormal on the interval  $[0, 1]$  with respect to the inner product  $(f, g) = \int_0^1 f(x)g(x)dx$ . Using these polynomials, find the best approximation in  $P^2[0, 1]$  for  $f(x) = x^{\frac{1}{2}}$ .

8. Consider the finite difference formula

$$f'(t_j) = \frac{1}{12h} [f(t_j - 2h) - 8f(t_j - h) + 8f(t_j + h) - f(t_j + 2h)] + O(h^4).$$

(a) Derive this formula by using Taylor's theorem.

(b) Derive this formula by using Lagrange polynomial representation.

9. Assume the numerical quadrature for  $\hat{f}(\hat{x})$  on  $[0, 1]$  is

$$\hat{J}(\hat{f}) = \int_0^1 \hat{f}(\hat{x})d\hat{x} \approx \hat{Q}(\hat{f}) = \sum_{j=0}^m \hat{\alpha}_j \hat{f}(\hat{x}_j).$$

Derive the numerical quadrature of  $J(f) = \int_a^b f(x) dx$ .

10. Consider numerically solving the initial value problem

$$y'(t) = f(t, y), \quad 0 < t \leq t_f, \quad \text{with } y(0) = \eta.$$

Assume  $f$  is sufficiently differentiable and let  $h$  denote the step size. Show that all convergent members of the family of methods

$$y_{n+2} + (\theta - 2)y_{n+1} + (1 - \theta)y_n = \frac{1}{4}h[(6 + \theta)f_{n+2} + 3(\theta - 2)f_n],$$

parameterized by  $\theta$ , are also  $A_0$ -stable.